

MLOps Pipeline for Tourism Package Prediction

Done by Mazin Alahmadi

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Executive Summary

- The tourism campaign has a 19.3% package-purchase rate, so better targeting can materially improve sales productivity.
- A completed MLOps pipeline now registers data, cleans and splits it, trains a tuned XGBoost model, and publishes a Hugging Face prediction app.
- The best local model achieved 91.3% test accuracy, 76.4% precision, 79.4% recall, and 77.8% F1 at the selected threshold.
- Business recommendation: prioritize high-probability customers first, then monitor conversion lift through the automated workflow.

Business Overview and Solution Approach

- Business problem: sales teams need to focus outreach on customers most likely to purchase a tourism package.
- Decision lens: rank prospects before calling so high-fit customers receive faster and better-timed follow-up.
- Solution: convert customer attributes into a purchase-probability score using a reproducible, deployment-ready ML workflow.
- Operating model: GitHub controls source code, Hugging Face stores data/model/app assets, and GitHub Actions runs the pipeline end to end.

Data Registration

- Master project folder: `tourism_project` with `data`, `model_building`, `deployment`, `hosting`, `reports`, and `models` subfolders.
- Raw data asset: `tourism_project/data/tourism.csv`, documented with a dataset README for governance.
- Hugging Face dataset target: `mazin903/tourism`
- Registration script creates or reuses the dataset repo and uploads the raw dataset files.
- Benefit: the workflow has a single governed data source instead of manual file movement.

Data Preparation

- Loaded the tourism dataset from Hugging Face when credentials are available, with local fallback for development.
- Removed generated index/customer ID fields, removed duplicates, and standardized inconsistent categorical values.
- Imputed missing numeric values with medians and categorical values with modes.
- Created a stratified 80/20 split: 3,208 training rows and 803 testing rows.
- Saved cleaned, train, and test files locally and prepared automatic upload back to the dataset repo.

Customer Insights That Drive Targeting

- Base purchase rate: 19.3%, confirming the need for selective campaign prioritization.
- Passport holders convert at 35.9% versus 12.4% for non-passport customers.
- Basic package prospects convert at 30.1%, the strongest product segment in the dataset.
- Single customers convert at 30.9%, more than double married and divorced segments.
- Recommendation: combine model scores with these high-conversion segments for call-list prioritization.

Model Building with Experimentation Tracking

- Model choice: XGBoost classifier with preprocessing for numeric and categorical fields.
- Class imbalance handled with `scale_pos_weight` so minority buyers are not ignored.
- Grid search evaluated 48 parameter/run combinations using F1 as the selection metric.
- MLflow logs parameters, metrics, artifacts, model path, and threshold analysis for comparison.
- Best model is packaged for Hugging Face Model Hub target: `mazin903/tourism-package-model`

Model Performance and Decision Threshold

- Selected threshold: 0.50, chosen for the best local F1 with a balanced precision/recall tradeoff.
- Test performance: 91.3% accuracy, 76.4% precision, 79.4% recall, 77.8% F1.
- Business meaning: most predicted buyers are actionable, while recall still captures nearly four out of five actual buyers.
- Top signals: Passport, Basic package, Single marital status, Executive designation, and CityTier.
- Operational use: score prospects, prioritize top-probability leads, and monitor uplift after deployment.

Model Deployment

- Deployment surface: Streamlit Community Cloud app, with Hugging Face hosting script and Space files retained for portability.
- The app loads the registered model artifact from Hugging Face Model Hub.
- User inputs are converted into the same feature schema used during model training.
- Output: purchase probability, decision threshold, and a business-friendly follow-up recommendation.
- Accepted deployment path: Streamlit app launched directly from the GitHub repository.

Automated GitHub Actions Workflow

- Trigger: push to main or manual workflow dispatch.
- Pipeline steps: install dependencies, register dataset, clean/split data, generate analysis, train/register model, and deploy app.
- Secrets and variables separate credentials from source code: HF_TOKEN, HF_USERNAME, dataset repo, model repo, and Space repo.
- Generated processed data and reports are committed back to main with skip-ci protection.
- Benefit: one repeatable workflow replaces manual notebook-only execution.

Output Evaluation Evidence

- GitHub repository: <https://github.com/mazin903/MLOps-pipeline-on-GitHub>
- Hugging Face dataset: <https://huggingface.co/datasets/mazin903/tourism>
- Hugging Face model: <https://huggingface.co/mazin903/tourism-package-model>
- Streamlit app deployment: Streamlit Community Cloud, using the GitHub repository and Hugging Face model.
- Hugging Face prediction app backup: <https://huggingface.co/spaces/mazin903/tourism-package-prediction>
- Screenshots to capture after first account run: repo folder structure, green GitHub Actions workflow, Hugging Face dataset/model pages, and prediction app result.

APPENDIX

Dataset Overview and Registration Details

- Source file: tourism.csv with customer demographics, interaction history, product pitch attributes, and purchase outcome.
- Target variable: ProdTaken, where 1 means the customer purchased the package and 0 means they did not.
- Model features include age, income, city tier, follow-ups, satisfaction, passport, car ownership, occupation, package pitched, and designation.
- Processed outputs: cleaned_tourism.csv, Xtrain.csv, Xtest.csv, ytrain.csv, and ytest.csv.
- Registration plan: raw and processed files are uploaded to the Hugging Face dataset repo for reproducible downstream steps.

Experimentation Tracking Details

- Algorithm: XGBoost classifier selected from the assignment-approved model family.
- Tuning grid: n_estimators 100/200, max_depth 3/5, learning_rate 0.05/0.10, reg_lambda 1.0/5.0, colsample values 0.8.
- Best parameters: n_estimators 200, max_depth 5, learning_rate 0.10, reg_lambda 1.0.
- Tracked artifacts: metrics, classification report, threshold analysis, feature importance, and joblib model bundle.
- Model Hub registration stores the trained model and model card for deployment reuse.

GitHub Actions Workflow Details

- Repository-ready workflow: `.github/workflows/pipeline.yml`.
- Step 1: install Python dependencies and validate the project environment.
- Step 2: register raw data, prepare train/test assets, and generate business summary tables.
- Step 3: train the model, log results, and upload the best bundle to Hugging Face Model Hub.
- Step 4: push deployment files to Hugging Face Spaces and commit generated reports back to main.

Evidence: GitHub Repository

Repository structure and project files: github.com/mazin903/MLOps-pipeline-on-GitHub

The screenshot displays the GitHub repository page for `mazin903/MLOps-pipeline-on-GitHub`. The repository is public and has 9 commits. The file list shows the following structure:

File	Description	Time
<code>.github/workflows</code>	Add free deployment path for project app	20 minutes ago
<code>outputs</code>	Add free deployment path for project app	20 minutes ago
<code>tourism_project</code>	Add Hugging Face dataset card metadata	2 minutes ago
<code>.env.example</code>	Add free deployment path for project app	20 minutes ago
<code>.gitignore</code>	Complete tourism MLOps pipeline	1 hour ago
<code>README.md</code>	Add free deployment path for project app	20 minutes ago
<code>low-code version.ipynb</code>	Add free deployment path for project app	20 minutes ago
<code>requirements.txt</code>	Add free deployment path for project app	20 minutes ago

The README file is titled "Tourism Package Prediction MLOps Pipeline" and describes the project's goal: to build an end-to-end MLOps pipeline for predicting customer purchase likelihood. The business goal is to prioritize customers with higher purchase likelihood, reduce manual targeting effort, and improve campaign conversion efficiency.

About

No description, website, or topics provided.

Releases

No releases published

Contributors 2

- mazin903
- github-actions[bot]

Languages

Language	Percentage
Python	56.5%
Jupyter Notebook	13.5%
JavaScript	10.8%
HTML	10.5%
CSS	8.1%
Dockerfile	0.6%

Evidence: GitHub Actions Workflow

Latest pipeline run #9 on commit 160aeda completed successfully on August 29, 2026.

The screenshot displays the GitHub Actions interface for a workflow named 'Tourism Project Pipeline' in the repository 'mazin903 / MLOps-pipeline-on-GitHub'. The specific workflow run is 'Fix Hugging Face dataset metadata #9', which has a status of 'Success' and a total duration of '1m 14s'. The interface includes a sidebar with navigation options: Summary, All jobs, Run details, Usage, and Workflow file. The main content area shows the workflow's execution details, including a table of jobs and a section for annotations. The 'mlops-pipeline' job is highlighted, showing its duration of '1m 10s'. The annotations section contains a warning about deprecated Node.js versions.

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← Tourism Project Pipeline

✓ **Fix Hugging Face dataset metadata #9** Sign in to view logs

Summary

All jobs

✓ mlops-pipeline

Run details

Usage

Workflow file

Triggered via push 2 minutes ago	Status	Total duration	Artifacts
mazin903 pushed to 160aeda main	Success	1m 14s	—

pipeline.yml
on: push

✓ mlops-pipeline 1m 10s

Annotations
1 warning

⚠ mlops-pipeline
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python.
[Show more](#)

Evidence: Hugging Face Dataset

Registered dataset repository: huggingface.co/datasets/mazin903/tourism

 **Hugging Face**

[Models](#) [Datasets](#) [Spaces](#) [Buckets](#) [NEW](#) [Docs](#) [Pricing](#) [Log In](#) [Sign Up](#)

Datasets:  mazin903: **tourism**  like 0

Tasks: [Tabular Classification](#) Languages: [English](#) Size: [1K< n < 10K](#) Tags: [tabular-classification](#) [tourism](#) [mlops](#) [sales-analytics](#)

[Dataset card](#) [Files and versions](#) [set](#) [Community](#)

Dataset Preview

Split (1)
train

The full dataset viewer is not available (click to read why). Only showing a preview of the rows.

Age float64	CityTier int64	DurationOfPitch float64	NumberOfPersonVisiting int64	NumberOfFollowups float64	PreferredPropertyStar float64	NumberOf float64
60	3	14	2	3		3
36	1	16	3	2		3
41	1	11	3	4		6
27	3	14	2	3		4
33	1	9	2	3		3
36	1	9	3	6		6

API Embed Duplicate Data Studio

Copy to bucket NEW

Downloads last month

Total file size:
1.3 MB

Tourism Package Prediction Dataset

This dataset contains customer profile and sales interaction records for Visit with Us. The target column is `ProdTaken`, where `1` means the customer purchased the offered package and `0` means the customer did not.

Files

Evidence: Hugging Face Model Hub

Registered model artifact repository: huggingface.co/mazin903/tourism-package-model

The screenshot displays the Hugging Face Model Hub interface for the 'tourism-package-model' by user 'mazin903'. The page is divided into several sections:

- Header:** Includes the Hugging Face logo, a search bar, and navigation links for Models, Datasets, Spaces, Buckets, Docs, Pricing, Log In, and Sign Up.
- Model Card:** The main section on the left, titled 'Tourism Package Prediction Model'. It contains a description: 'This model predicts whether a customer is likely to purchase the Wellness Tourism Package.' Below this, there are sections for 'Model' and 'Training'.
 - Model:** Lists details such as the algorithm (XGBoost binary classifier), preprocessing steps (numeric median imputation, standard scaling, categorical mode imputation, one-hot encoding), the target variable (ProdTaken), and the decision use (prioritize sales outreach and campaign targeting).
 - Training:** Describes the training process, including stratified train/test splitting, class imbalance weighting, grid-search hyperparameter tuning, threshold analysis, and MLflow experiment tracking.
- Evaluation:** A section titled 'Evaluation' showing local validation generated test metrics:
 - Accuracy: 0.913
 - Precision: 0.764
- Right Sidebar:** Contains a 'Downloads last month' bar chart and a section for 'Inference Providers'. It lists 'Tabular Classification' as the provider and includes a message: 'This model isn't deployed by any Inference Provider.' with a link to 'Ask for provider support'.

Evidence: Streamlit Community Cloud App

Primary accepted app deployment: tourism-package-prediction-mazin903.streamlit.app

Streamlit - Tourism Package Prediction - Streamlit

Share ☆ ⚙️ 🔄 Deploy ⋮

Tourism Package Prediction

Estimate whether a customer is likely to purchase the Wellness Tourism Package.

Age

18 35 70

Preferred property star

3

Type of contact

Self Enquiry

Marital status

Single

City tier

1

Annual number of trips

1 3 22

Duration of pitch (minutes)

5 15 130

Has passport?

Yes

Occupation

Salaried

Pitch satisfaction score

1 3 5

Gender

Male

Owns a car?

Yes

Number of people visiting

Number of children visiting

< Manage app

Evidence: Hugging Face Static Space Backup

Free hosted prediction-app backup: huggingface.co/spaces/mazin903/tourism-package-prediction

Spaces

mazin903 / tourism-package-prediction

like 0

Running

App

Files

Community

VISIT WITH US

[GitHub](#) [Dataset](#) [Model](#)

Tourism Package Prediction

LEAD SCORING

Age

35

Monthly income

23000

City tier

Tier 1

Duration of pitch

15

People visiting

3

Follow-ups

4

Preferred property star

3 stars

Annual trips

3

Passport

No

Pitch satisfaction

3

Owns a car

Yes

Children visiting

1

Type of contact

Self Enquiry

Occupation

Salaried

Gender

Male

Product pitched

Basic

Marital status

Single

Designation

Executive

Customer profile

Model loaded from Hugging Face Model Hub

23.2%

purchase probability

Recommendation: place this customer in a lower-touch nurture segment.

Test accuracy

91.8%

Test F1

0.792

Threshold

0.45

Top model drivers

Passport

Designation Executive

ProductPitched Basic

MaritalStatus Single

CityTier